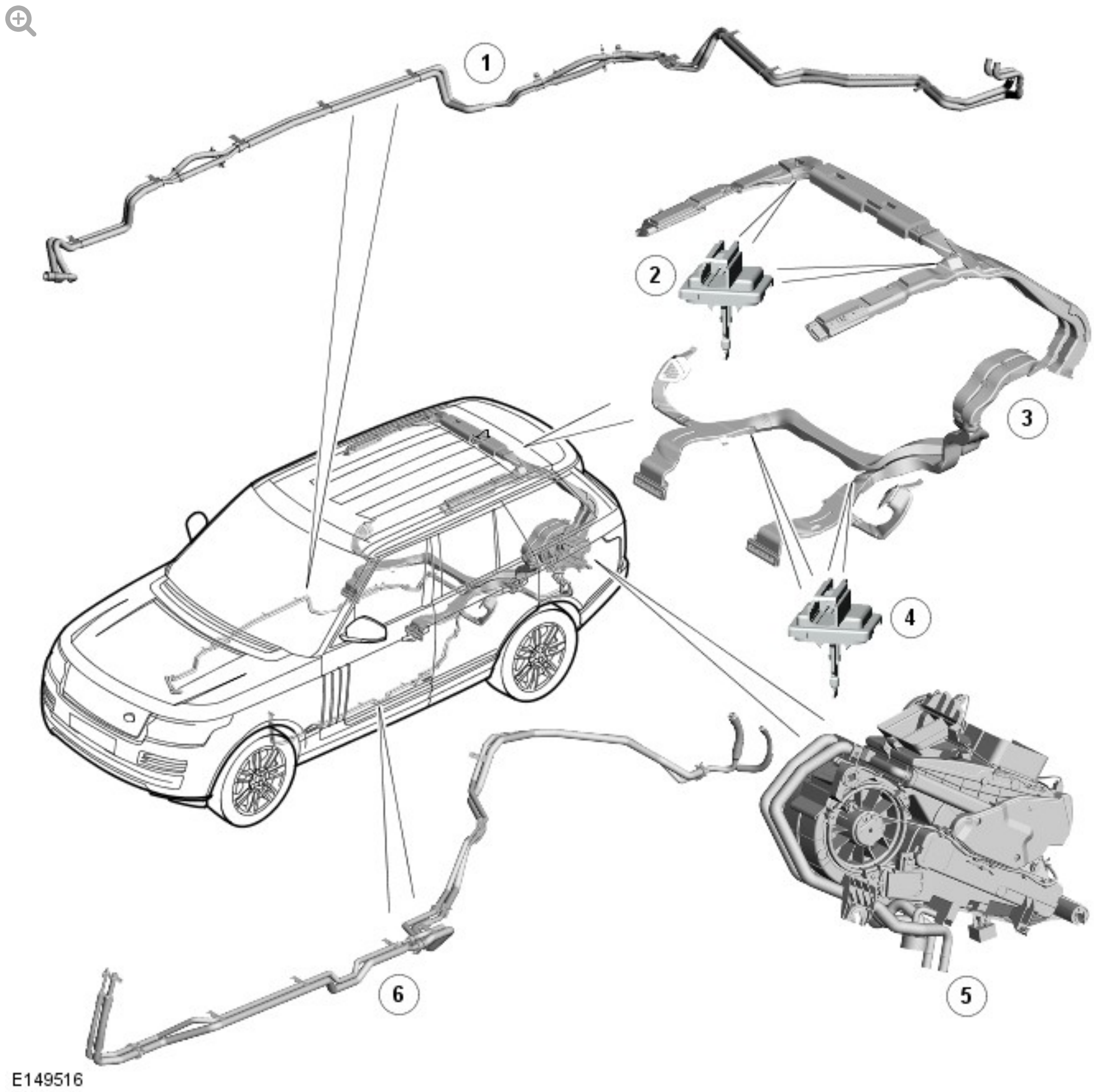
2016.0 RANGE ROVER (LG), 412-02

AUXILIARY CLIMATE CONTROL

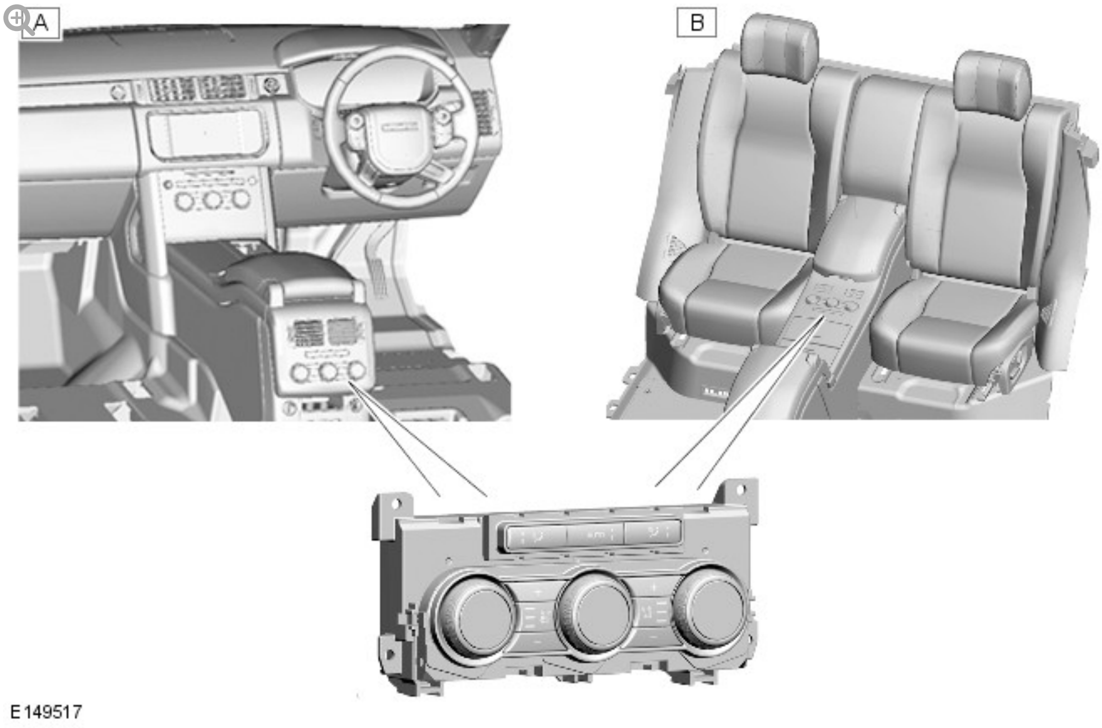
DESCRIPTION AND OPERATION

COMPONENT LOCATION - SHEET 1 OF 2



ITEM	DESCRIPTION
1	Coolant pipes
2	Duct air temperature sensor - face ducts
3	Air distribution system
4	Duct air temperature sensor - lap and foot ducts
5	Auxiliary climate control assembly
6	Air conditioning pipes

COMPONENT LOCATION – SHEET 2 OF 2



ITEM	DESCRIPTION
A	Rear integrated control panel - all except business class seats
B	Rear integrated control panel – business class seats

OVERVIEW

The auxiliary climate control system provides additional air distribution vents and separate temperature zones for the left and right areas of the rear seats. The auxiliary climate control system consists of:

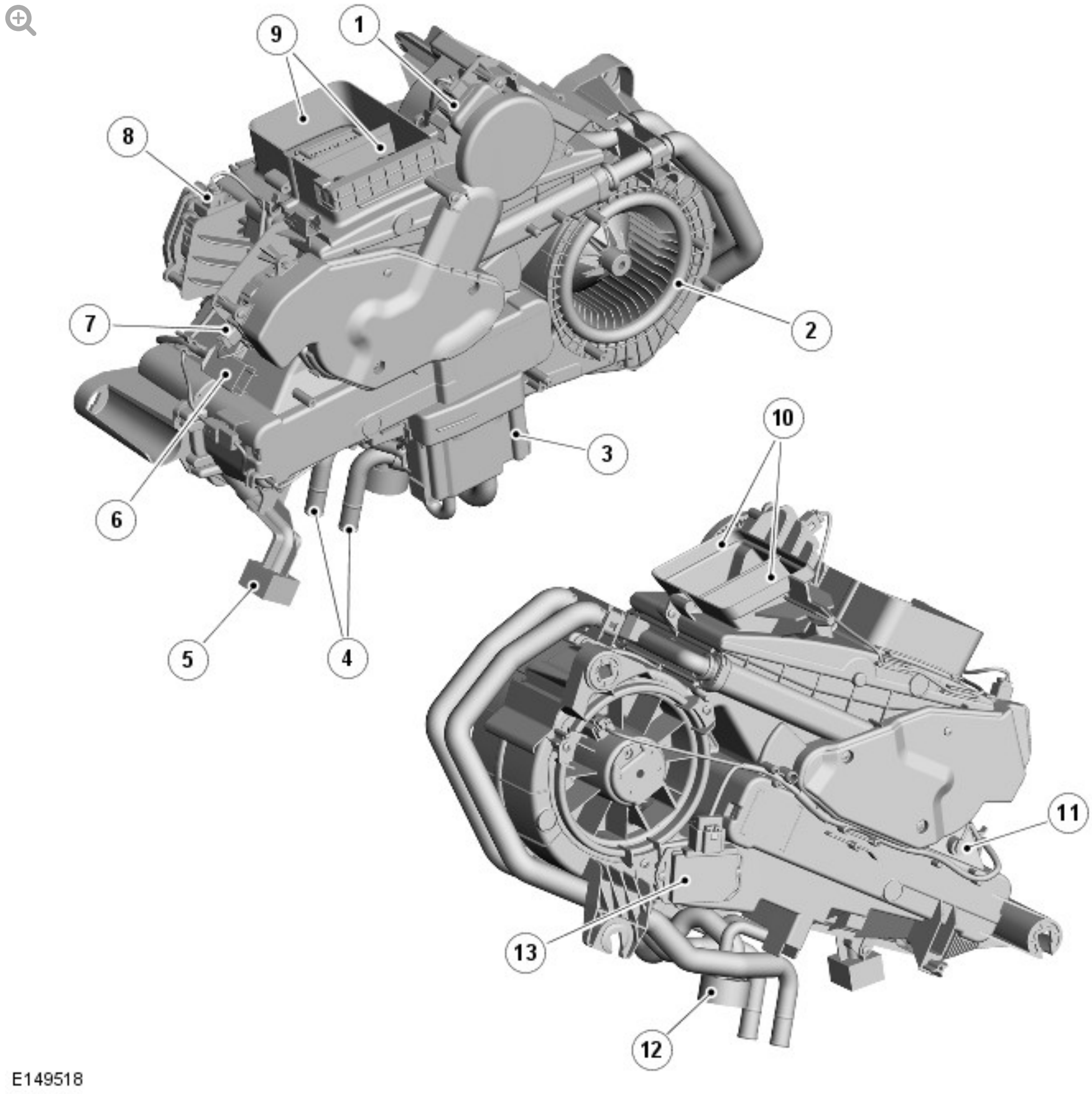
- An auxiliary climate control assembly.
- A refrigerant circuit.
- A heating circuit.
- A distribution system.
- The rear ICP (integrated control panel).

Operation of the auxiliary climate control system is controlled by the ATCM (automatic temperature control module), the rear integrated control panel and the touch screen.
For additional information, refer to: Control Components (412-01B Climate Control - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

Air from the passenger compartment is recirculated through the auxiliary climate control assembly, where the air is temperature regulated and then directed through the distribution system to vents in the headliner, the rear seat side bolsters and the carpet in front of the rear seats.

DESCRIPTION

AUXILIARY CLIMATE CONTROL ASSEMBLY



ITEM	DESCRIPTION
1	Distribution motor
2	Blower
3	Isolation valve
4	Coolant pipes
5	Evaporator drain tube
6	Electrical connector for vehicle harness
7	Right temperature blend motor
8	Left temperature blend motor
9	Face distribution outlets (to headliner vents)
10	Lap and foot level distribution outlets (to seat side bolster and carpet vents)

11	Evaporator temperature sensor
12	Air conditioning pipe connector block
13	Blower control module

The auxiliary climate control assembly is installed on the left side of the luggage compartment, behind the rear quarter trim panel. A grille in the trim panel allows air to flow from the luggage compartment into the auxiliary climate control assembly.

The auxiliary climate control assembly consists of a casing, formed from a pair of plastic moldings, which contains:

- An evaporator
- A heater core
- Temperature blend doors
- A distribution door
- A blower
- A blower control module
- An evaporator temperature sensor.

Refrigerant and coolant pipes from the engine compartment are connected to pipes from the evaporator and the heater core immediately below a seal plate in the rear left wheel arch. An evaporator drain tube, sealed with a foam rubber gasket, locates in the top of the left side member of the rear floor.

In the casing of the auxiliary climate control assembly, two parallel air steams flow from the blower through the evaporator and heater core to two pairs of distribution outlets. The left and right air streams supply the related rear zones of the passenger compartment. The front pair of distribution outlets supply the vents in the seat side bolsters and the carpet. The rear pair of distribution outlets supply the vents in the headliner.

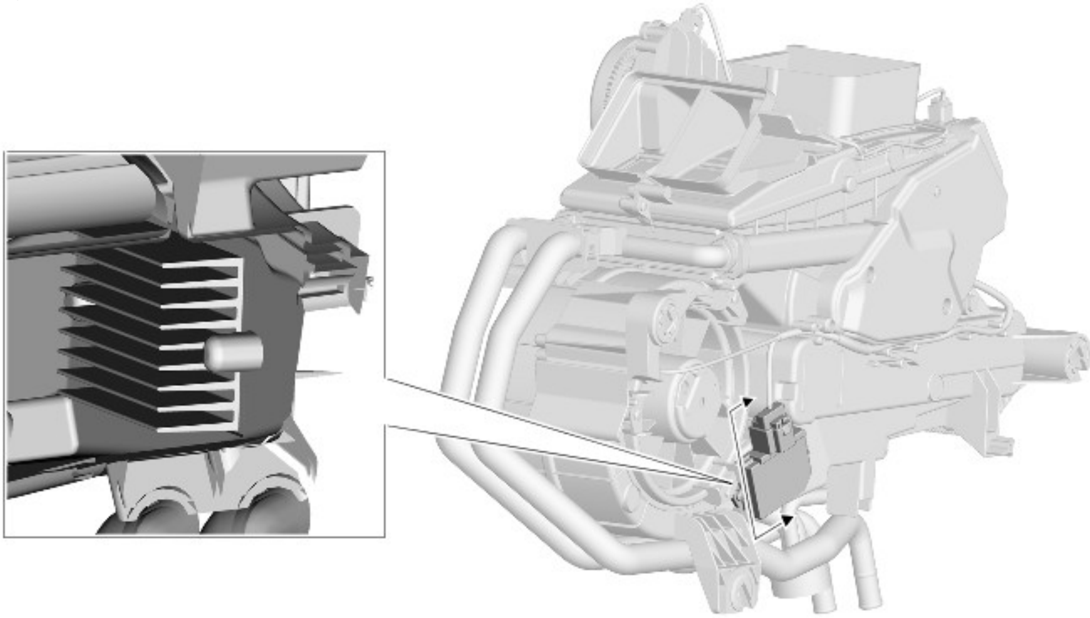
The left and right air streams each contain two temperature blend doors connected together with a link mechanism. Separate electric motors, installed on the rear of the casing, operate the temperature blend doors in each air stream to give independent temperature control for each rear climate zone.

The left and right air streams share a common distribution door installed below the distribution outlets. An electric motor, installed on the top of the casing, operates the distribution door.

A LIN (local interconnect network) bus connects the distribution motor and the temperature blend motors in series with the ATC (automatic temperature control) module. The motors are powered by feeds from the ATC module, and share a ground connection with the rear blower and rear blower control module.

The ATCM (automic temperature control module) determines the positions of the distribution and temperature blend doors by using either their closed or open position as a datum and memorizing the steps that it drives the individual motors. Each time the ATCM (automic temperature control module) is activated, it checks the memorized position of the motors against fixed values for the current distribution and temperature settings on the control panel. If there is an error, the ATCM (automic temperature control module) calibrates the applicable motor, to re-establish the datums, by driving it fully closed or open before re-setting it to the nominal selected position. A calibration run can also be invoked using Land Rover approved diagnostic equipment.

BLOWER CONTROL MODULE



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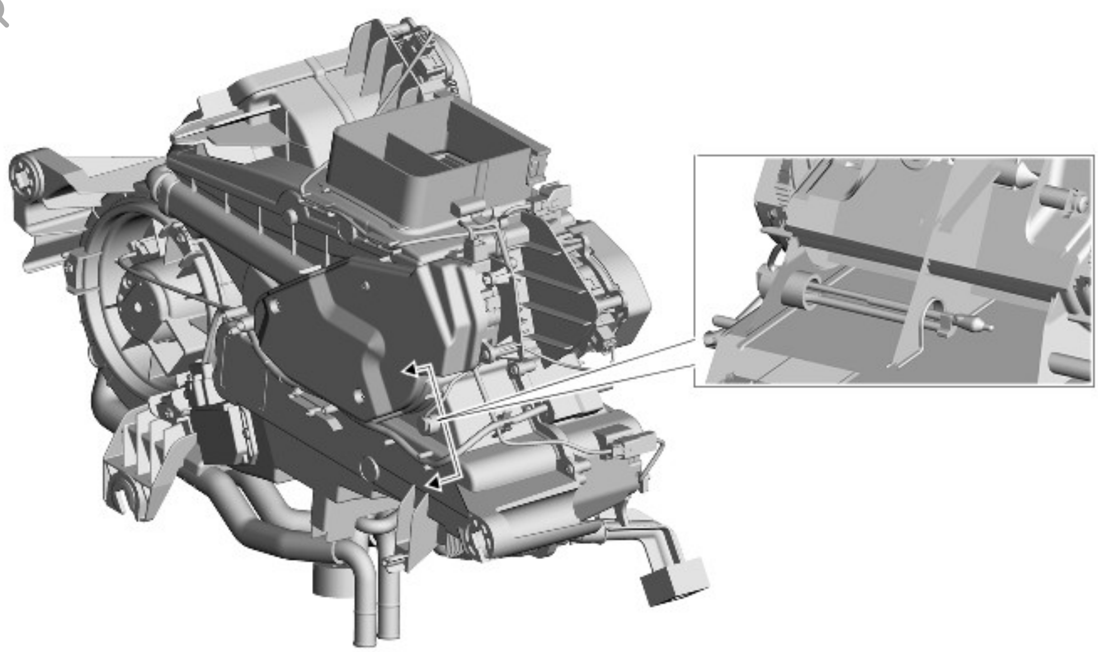
The blower consists of an open hub, centrifugal fan powered by an electric motor. Power for the blower is supplied from the rear blower relay in the EJB (engine junction box) via the blower control module, which is installed in the left side of the auxiliary climate control assembly, immediately downstream of the blower. To control blower speed, the ATC module outputs a PWM (pulse width modulation) signal to the blower control module to regulate the voltage of the blower power supply.

EVAPORATOR TEMPERATURE SENSOR



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The evaporator temperature sensor is installed in the auxiliary climate control assembly on the downstream side of the evaporator. The evaporator temperature sensor is a NTC (negative temperature coefficient) thermistor connected to the rear integrated control panel.

REFRIGERANT CIRCUIT

Low and high pressure pipes connect the front A/C (air conditioning) system to the auxiliary climate control assembly at a connector block in the left rear wheel arch, behind the wheel arch liner. On the auxiliary climate control assembly, pipes from the connector block are attached to an isolation valve on the evaporator.

AIR CONDITIONING PIPES

The A/C pipes consist of three sections of insulated pipe. The front and rear sections incorporate hoses at the ends that connect to the front A/C system and the auxiliary climate control assembly respectively. From the auxiliary climate control assembly, the A/C pipes are routed forward around the inboard side of the left rear wheel arch, then under the left edge of the floorpan to the rear of the left front wheel arch. The A/C pipes connect to the front climate control A/C system at a grommet in the wheel arch.

ISOLATION VALVE

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The isolation valve is a combined solenoid operated shut-off valve and block-type thermostatic expansion valve attached to the inlet and outlet pipes of the evaporator. The ATC module controls the operation of the shut-off valve to isolate the auxiliary climate control assembly from the front A/C system.

HEATER CIRCUIT

Supply and return coolant pipes connect the engine cooling system to the heater core pipes of the auxiliary climate control assembly in the left rear wheel arch, behind the wheel arch liner.

The coolant pipes consist of sections of insulated rigid pipes connected together with flexible hose joints. Quick release fittings connect the coolant pipes to the engine cooling system.

From the auxiliary climate control assembly, the coolant pipes are routed along the rear cross member and then forward under the right edge of the floorpan to the chassis longitudinal at the rear of the right front wheel arch liner.

DISTRIBUTION SYSTEM

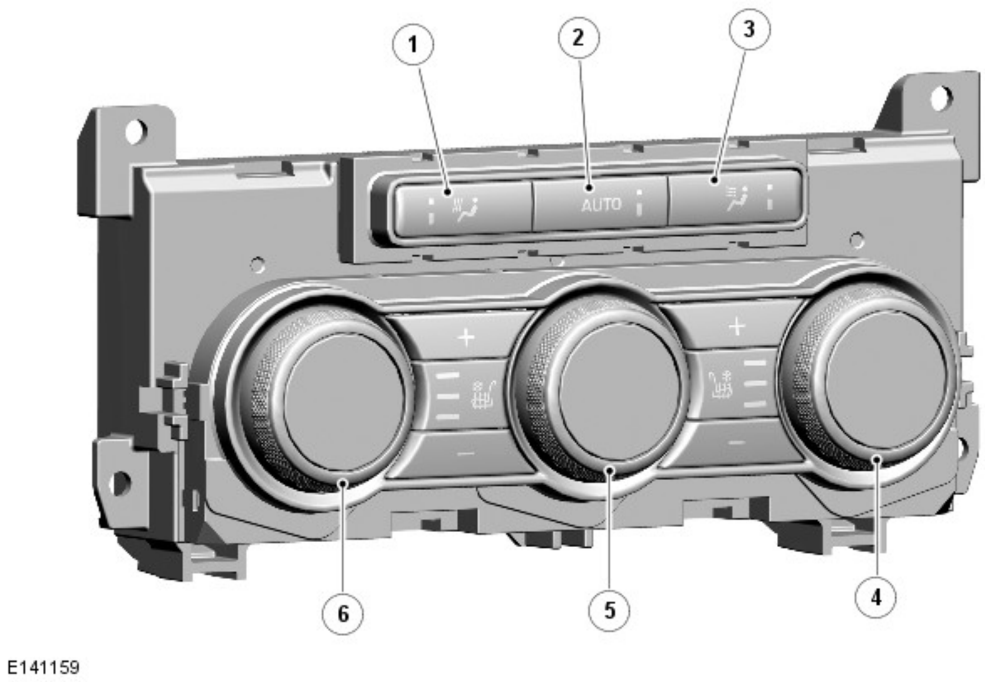
The distribution system consists of a network of air ducts that supply air from the distribution outlets of the auxiliary climate control assembly to the vents installed in:

- The headliner, above each rear door
- The left and right rear seat side bolsters
- The carpet, below the left and right rear seats.

The vents in the headliner are adjustable, those in the side bolsters and the carpet are fixed.

Four duct air temperature sensors are installed in the air ducts. One in each duct to the left and right vents in the headliner, one in the duct to the left vents in the seat side bolsters and carpet, and one in the duct to the right vents in the seat side bolsters and carpet. The sensors are NTC thermistors, which are connected to the rear integrated control panel. The ATCM (automic temperature control module) uses the data from the sensors in output temperature calculations for the related ducts.

REAR INTEGRATED CONTROL PANEL



ITEM	DESCRIPTION
1	Foot distribution switch
2	Automatic mode switch

3	Face distribution switch
4	Right temperature switch
5	Blower switch
6	Left temperature switch

The rear ICP is installed either in the rear of the floor console or, on vehicles with business class seats, in the rear seat console. The control panel contains rotary and push button switches for controlling the temperature, volume and distribution of air ro the rear seats. Each push switch contains an amber status LED (light emitting diode). Display screens are integrated into the center of the rotary switches to display the selected temperature and blower speed as appropriate.

Additional push switches on the rear ICP are provided for control of seat heating/climate control seats.
For additional information, refer to: Seats (501-10, Description and Operation).

The rear ICP has a permanent battery power feed from the CJB (central junction box), and is connected to the medium speed CAN (controller area network) comfort bus. The rear ICP converts selections on the climate control switches into medium speed CAN comfort bus messages and transmits them to the ATCM (automic temperature control module). The rear ICP also receives inputs from the auxiliary system evaporator temperature sensor and four duct air temperature sensors. These inputs are also converted into medium speed CAN comfort bus messages and transmitted to the ATCM (automatic temperature control module).

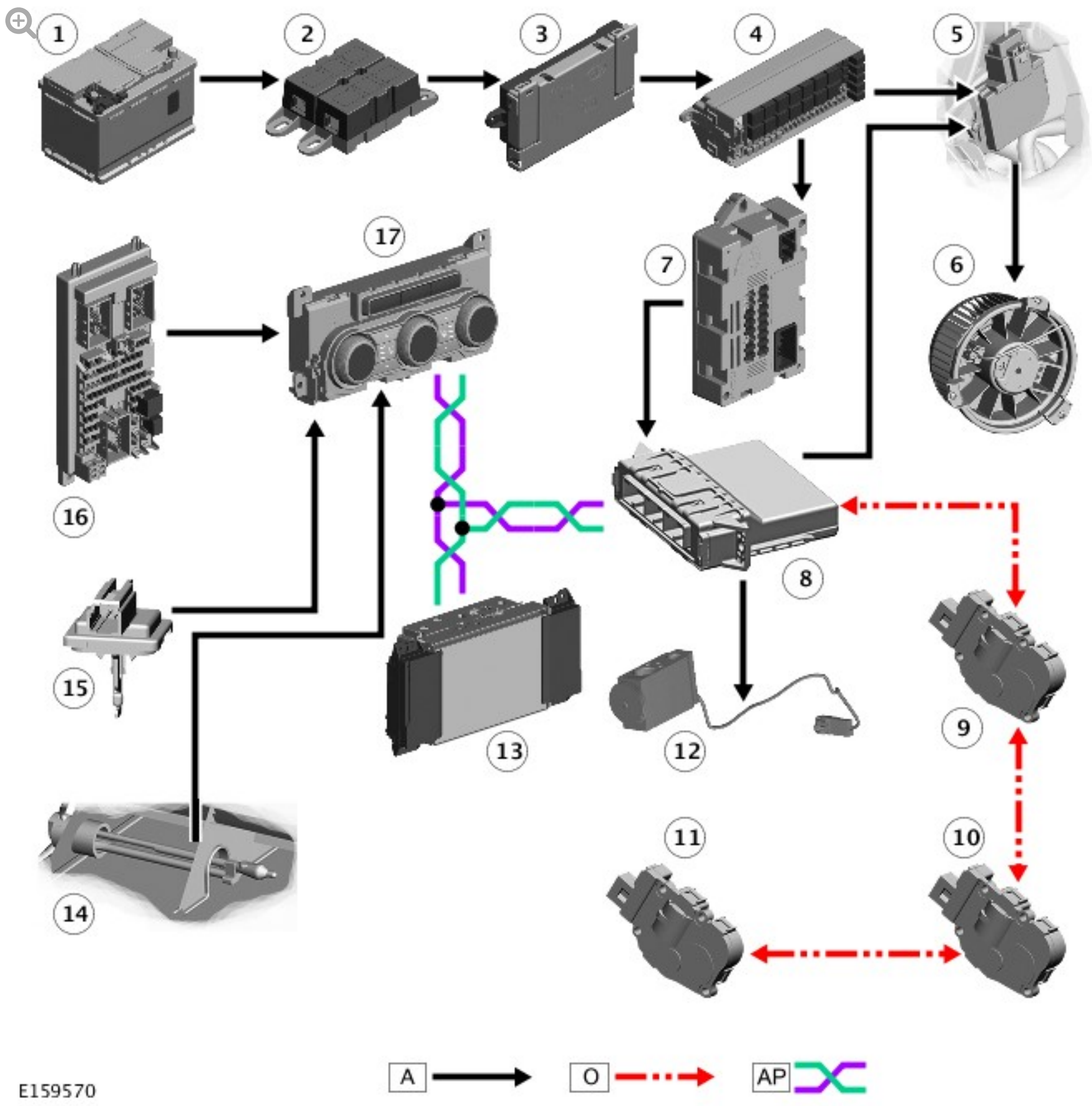
OPERATION

For the auxiliary climate control system to operate, the engine must be running and automatic mode selected on the rear ICP or the **Rear climate** menu of the touch screen. The ATCM (automic temperature control module) then illuminates the LED in the rear ICP AUTO switch, energizes the isolation valve and the rear blower, and operates the temperature blend and distribution motors to maintain the temperatures selected on the rear ICP. If the main climate control system is not already operating, the ATCM (automic temperature control module) also activates the A/C compressor and operates the main climate control system in automatic mode.

Once the auxiliary climate control system is operating, the blower, distribution and temperature can be adjusted as required on the rear ICP or the **Rear climate** menu of the touch screen. More than one distribution setting at a time may be selected to achieve the required air distribution.

The auxiliary climate control system can be selected off on the rear ICP, using the AUTO switch, or on the **Rear climate** menu of the touch screen. The **Rear climate** menu can also be used to lock the rear ICP to prevent operation of the auxiliary climate control system from the rear seats.

CONTROL DIAGRAM



A = HARDWIRED; O = LIN BUS; MS (MEDIUM SPEED) CAN COMFORT BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Blower control module
6	Blower
7	Quiescent Current Control Module (QCCM)
8	Automatic Temperature Control Module (ATCM)

9	Left temperature blend motor
10	Right temperature blend motor
11	Distribution motor
12	Isolation valve
13	Touch Screen (TS)
14	Evaporator temperature sensor
15	Duct air temperature sensor (4 off)
16	Central Junction Box (CJB)
17	Rear Integrated Control Panel (RICP)